SAFETY SIGNAL REVIEW CAPSTONE

Pharmacovigilance | Descriptive review of a synthetic adverse-reaction dataset

Project purpose
Summarize reported case characteristics, seriousness-related information, outcomes, and reaction-onset timing; identify a cautious potential safety pattern; and propose follow-up evidence needs without claiming causation.

Project detail	Value
Dataset	adverse_drug_reactions_sample.csv
Records reviewed	8
Data type	Synthetic training data; not for real safety reporting
Prepared	23 October 2026
Scope	Descriptive review and hypothesis generation only

Important limitation: this is a small, selected set of eight reports. It has no denominator, comparator group, verified medical records, exposure details, or reporting dates. The findings cannot estimate incidence or establish causality.

1. Executive Summary

This review covers **8 synthetic reports** involving eight different suspected products. Three reports recorded hospitalization as an outcome, and one report was labelled life-threatening. These fields describe the supplied records; the dataset does not include a separate formal seriousness assessment field, so outcome and severity must not be treated as interchangeable.

Potential pattern for follow-up: two reports describe bleeding-related reactions—“Gastric Ulcer Bleed” with ibuprofen (ADR-9004) and “Uncontrolled Hemorrhage” with warfarin (ADR-9007). Both are clinically important descriptions in the supplied data, but they involve different products and indications. This is a hypothesis-generating observation, not evidence that either product caused the event or that a shared signal has been established.

Key findings

Measure	Observed result	Interpretation
Reports	8	Small descriptive sample
Severity labels	3 mild; 2 moderate; 2 severe; 1 life-threatening	Severity reflects recorded intensity label
Outcomes	3 recovered; 1 recovering; 3 hospitalized; 1 not recovered	Outcome categories are not mutually equivalent to seriousness
Reaction onset	1–30 days; median 8.5 days	Onset timing is not a reporting trend
Potential pattern	2 bleeding-related reports	Needs case-level verification and context

Overall conclusion: prioritize case-level review of the two bleeding-related reports and the life-threatening warfarin report, while retaining a non-causal interpretation. The current dataset is insufficient to confirm a safety signal.

2. Dataset and Method

Source: the user-provided synthetic CSV, containing ReportID, PatientAge, PatientGender, SuspectedDrug, Indication, AdverseReaction, ReactionOnsetDays, Severity, Outcome, and ReporterType. All eight rows were included. No rows were removed or imputed.

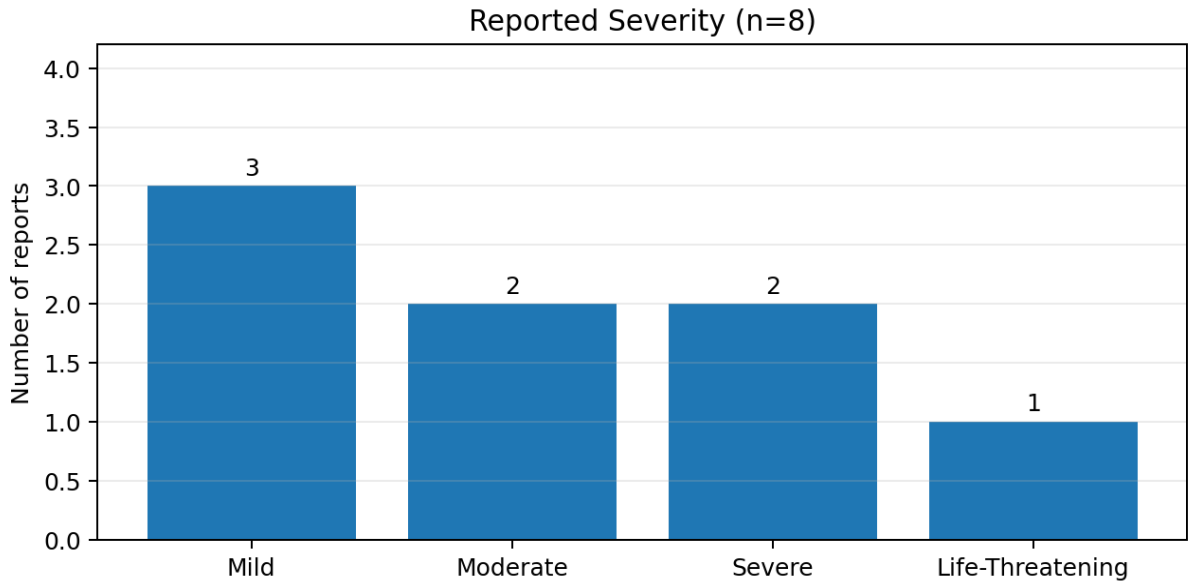
Method: simple descriptive counts, proportions, range/median of reaction-onset days, and review of reaction terms for a possible recurring clinical theme. Percentages use n=8 as the denominator and are descriptive only. No statistical testing, disproportionality analysis, causal inference, or incidence estimation was performed.

Data-quality check

Check	Result
Rows / columns	8 rows / 10 columns
Missing values in supplied CSV	0 cells missing
Unique report IDs	8 of 8
Unique suspected products	8 of 8
Dates for reporting trend	Not supplied
Exposure denominator / comparator	Not supplied

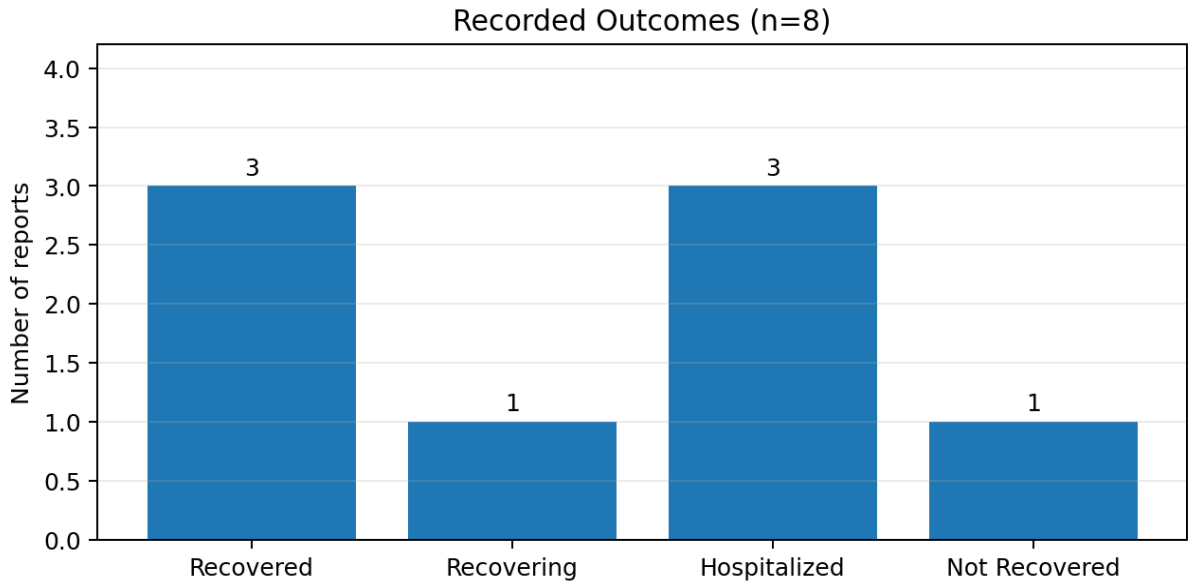
3. Descriptive Findings

Severity labels



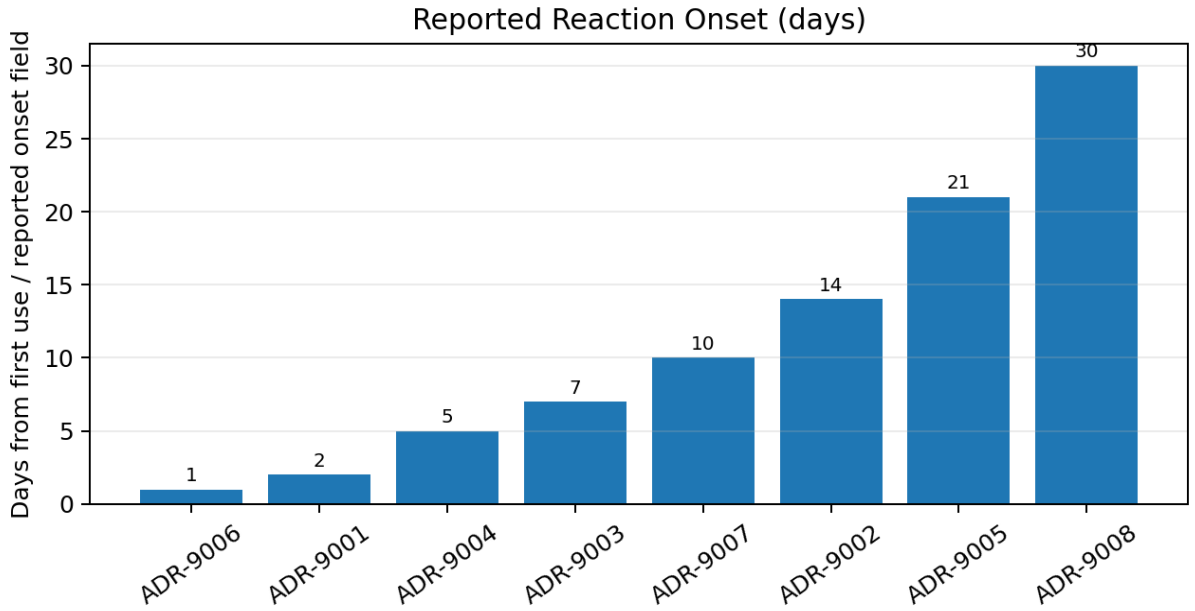
Three reports were labelled mild, two moderate, two severe, and one life-threatening. A life-threatening label is important for review, but severity and regulatory seriousness are distinct concepts.

Recorded outcomes



Three reports recorded hospitalization, three recovery, one recovering, and one not recovered. The dataset does not provide a formal seriousness field or sufficient detail to independently validate every seriousness criterion.

4. Reaction-Onset Timing and Case Listing



Reaction-onset values range from 1 to 30 days; median 8.5 days. These values describe reported onset timing only. Because no report dates or longitudinal time series are supplied, a calendar reporting trend cannot be assessed.

Case-level data

Report ID	Suspected drug	Reaction	Onset (days)	Severity	Outcome
ADR-9001	Amoxicillin	Skin Rash & Hives	2	Mild	Recovered
ADR-9002	Atorvastatin	Severe Muscle Myalgia	14	Moderate	Recovering
ADR-9003	Ciprofloxacin	Tendonitis	7	Severe	Hospitalized
ADR-9004	Ibuprofen	Gastric Ulcer Bleed	5	Severe	Hospitalized
ADR-9005	Lisinopril	Persistent Dry Cough	21	Mild	Recovered
ADR-9006	Metformin	Nausea & Diarrhea	1	Mild	Recovered
ADR-9007	Warfarin	Uncontrolled Hemorrhage	10	Life-Threatening	Hospitalized
ADR-9008	Paclitaxel	Peripheral Neuropathy	30	Moderate	Not Recovered

Case listing reproduces the supplied fields; it does not independently validate diagnoses, product exposure, or event causality.

5. Potential Signal Assessment

Observation: two reports contain bleeding-related event terms: ADR-9004 (ibuprofen; “Gastric Ulcer Bleed”; severe; hospitalized) and ADR-9007 (warfarin; “Uncontrolled Hemorrhage”; life-threatening; hospitalized). The latter is the highest-priority individual record based on the supplied severity and outcome fields.

Why this is not a confirmed signal

• The records involve different suspected products and different indications.
• There are only two bleeding-related reports in this selected eight-record sample.
• No exposed-population denominator, comparator, background rate, or reporting-period information is available.
• Confounding factors, concomitant medicines, dose, duration, medical history, laboratory results, dechallenge/rechallenge, and clinical confirmation are not supplied.
• The field “SuspectedDrug” is a dataset label and does not establish causality.

Alternative explanations to investigate

• Underlying condition or comorbidities may contribute to the event.
• Concomitant products or interactions may be relevant.
• Dose, duration, adherence, overdose, or medication-use error may be relevant.
• The reported event term may need clarification or clinical confirmation.
• Selection and reporting effects may explain why these cases appear in this small sample.

Provisional assessment

Dimension	Assessment
Potential signal	Bleeding-related reports warrant follow-up review
Evidence strength	Very limited; descriptive observation only
Causality	Cannot be determined from this dataset
Recommended status	Hypothesis for further evaluation; not confirmed

This assessment is an exercise in signal triage, not a clinical or regulatory conclusion. No treatment advice is provided.

6. Follow-up Evidence and Questions

Priority	Information to request	Purpose
High	Confirm event details, diagnosis, dates, clinical course, and outcome for AD-R-0007 and AD-R-0004.	Verify the reported bleeding events.
High	Relevant medical history, indication details, and prior bleeding or gastrointestinal history.	Assess alternative explanations.
High	All concomitant medicines, OTC products, supplements, and recent changes.	Explore possible interactions or co-exposures.
High	Product dose, route, start/stop dates, adherence, medication errors, and any exposure information.	Clarify exposure and temporal relationship.
High	Relevant laboratory results and clinical findings, if available and appropriate to disclose.	Support case interpretation; do not infer missing results.
Medium	Dechallenge/rechallenge information, including whether the product was stopped or restarted and what happened.	Classify the role of exposure and what remains unknown.
Medium	Reporter/source details sufficient for follow-up under applicable privacy and reporting processes.	Improve completeness while protecting personal data.

Risk-minimization questions

1. Do the verified case details suggest any preventable medication-use factors?
2. Are current product information and local risk communications appropriate for the verified evidence?
3. Is targeted education or a prescribing/dispensing safeguard justified after expert review?
4. What measurable outcome would show whether a risk-minimization action is working (for example, completeness of follow-up, recurrence patterns, or process compliance)?

Any risk-minimization action should be based on verified evidence, relevant product information, and qualified pharmacovigilance/medical review—not on this training sample alone.

7. Limitations and Responsible Interpretation

• Synthetic, very small dataset (n=8); not representative of a real population.
• No exposure denominator, comparator group, or background event rates.
• No report dates; reporting trends over time cannot be assessed.
• No separate seriousness field; hospitalization is an outcome label, not a complete seriousness assessment by itself.
• No case narratives, source documents, concomitant medicines, laboratory data, dechallenge/rechallenge, or verified diagnoses.
• One report per suspected product means product-level comparisons are not meaningful.
• Descriptive counts cannot establish causality, incidence, relative risk, or a confirmed safety signal.

8. Conclusion

The eight-record synthetic dataset contains three hospitalization outcomes and one life-threatening severity label. Two reports describe bleeding-related events and merit focused case-level follow-up as a hypothesis-generating pattern. However, the sample is too small and lacks the clinical, exposure, comparator, and reporting-time information needed to confirm a signal or infer causation.

Recommended next step: verify the two bleeding-related case records, obtain the missing clinical and medication-use context, and reassess only after evidence is reviewed. Keep all conclusions explicitly provisional.

9. Resume-Ready Project Description

Pharmacovigilance Practice Project | Safety Signal Review
Self-directed training project using synthetic data

• Reviewed an 8-record synthetic adverse-reaction dataset and summarized severity labels, outcomes, and reaction-onset timing using descriptive analysis.
• Built charts and case-level tables to communicate safety-data patterns and data limitations.
• Flagged two bleeding-related reports for follow-up assessment while documenting alternative explanations and missing evidence.
• Applied cautious pharmacovigilance reasoning by distinguishing severity from seriousness and avoiding unsupported causality claims.

Suggested skills / ATS keywords

Pharmacovigilance • Adverse Event Review • Safety Signal Detection (training) • Case Triage • Seriousness vs Severity • Descriptive Data Analysis • Safety Data Interpretation • Follow-up Questions • Risk Minimization • Data Quality Review • Report Writing

10. References and Scope Note

Primary task source: EdVyro “Safety Signal Review Capstone” brief and the supplied adverse-reaction aggregate CSV. The project brief describes the dataset as synthetic aggregate training data. This report does not use real patient data and is not medical advice or a real safety report.

Learning references listed in the EdVyro task library:
• ICH E2A: Clinical Safety Data Management — https://database.ich.org/sites/default/files/E2A_Guideline.pdf
• WHO–UMC causality assessment system — https://who-umc.org/media/u4gjgxvv/who-umc-causality-assessment_new-logo.pdf
• FDA: What is a Serious Adverse Event? — https://www.fda.gov/safety/reporting-serious-problems-fda/what-serious-adverse-event

End of report • Synthetic training data only